Ky Heon

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Education

University of California, Los Angeles (UCLA)

B.S. in Mechanical Engineering June 2023

Relevant Coursework:

- Control System Design, Kinematics of Robotics, Numerical Methods & Analysis, Algorithms, Fourier Optics

Experience

SCADA & Systems Engineer August 2025 – Current

Divcon Controls *Dallas, TX*

- Designed and commissioned Ignition SCADA systems for hyperscaler AI compute infrastructure across the US
- Developed an AWS Kinesis Streams pipeline delivering real-time KPI dashboards for datacenter SLA compliance
- Built an Ignition graphics library, eliminating subcontractor reliance and reshoring $500,000 in annual expenditure
- Coded a Python CLI to automate Modbus/BACnet device integration, eliminating 95% of manual configuration
- Introduced Git workflows to the OT environment, standardizing version control across the engineering team and accelerating project delivery

Automation & Controls Engineer June 2024 – August 2025

Freezer Engineering *Denver, CO*

- Designed PlantPAx and Ignition SCADA systems for projects across the oil, gas, chemical & mining industries
- Independently developed distributed control infrastructure to manage facility safety and optimize production
- Coordinated with engineers in the development of network architectures, one-line diagrams, and arc flash studies
- Managed client relations and led cross functional teams to execute on projects generating over $2 million
- Designed a data-driven model predictive control algorithm reducing annual operational costs by over $150,000

Software Developer September 2021 – July 2023

ASME X1 Robotics *Los Angeles, CA*

- Built a video streaming server to host real-time camera feed from BruinBot using Flask and Ngrok
- Pioneered the development of the computer vision algorithm allowing "BruinBot" to detect human features, enhancing its abilities to interpret and intelligently respond to its environment
- Researched and implemented cloud computing strategies to offload the computer vision to a serverless environment, decreasing tracking latency by over 80%
- Solved the forward and inverse kinematics of the quadruped robot dog "Boelt"

Projects

HandAid

- Created a 4 DOF serial robot to help hospital patients with limited mobility feed themselves
- Coded a Julia package to solve the forward and inverse kinematics of the 4R manipulator
- Simulated the pathing of the robot and implemented the kinematics with servo motors and a PS4 controller

Secure AI

- Developed a platform that lets users search through video footage using natural language
- Designed a RAG based multi-modal machine learning engine that reduced model hallucinations by over 90%
- Created demos and presentations of the novel platform to silicon valley venture capitalists
- Winner of LlamaIndex's 2024 Machine Learning Hackathon with over 400 participants

Skills

CAD & FEA: SOLIDWORKS, Autodesk Inventor, AutoCad, Plant3D, Fusion 360

Languages: Python, MATLAB, Julia, TS, C/C++, Java

Automation: Ignition, PlantPAx, TIA, Studio 5000, VFD (Eaton, Allen-Bradley, Schneider, Danfoss)

Communications: Modbus, BACnet, OPC-UA, Profibus, I2C, IO Link, ProSoft, Red Lion

Tools: Pavilion8 MPC, Kepware, AWS, SIMULINK, React, Flask, Linux, Excel, LaTeX

Other: Strong skills in leadership and technical communication